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Thinking and Programming
Semester- odd
Batch- ALL

Tutorial Assignment 7

Tutorial title: Practicing Strings, Pattern matching and Functions

CO Mapping

Name	CO1	CO2	CO3
Loops, Sequence Operations	✓	-	-

Strings

- String literals in python are surrounded by either single quotation marks, or double quotation marks. Ex. 'hello' is the same as "hello".
- You can display a string literal with the `print()` function:

Example:
`print("Hello")`
`print('Hello')`

Output:
Hello
Hello

- Assigning a string to a variable is done with the variable name followed by an equal sign and the string. For multi line string use three double quotes or three single quotes.

Single Word or single line string
`a = "Hello"`
`print(a)`

Assigning Multi line string
`a = """ Multi line string can be assigned
using three double quotes, or
three single quotes"""`
`print(a)`

- Like many other popular programming languages, strings in Python are arrays of bytes representing Unicode characters.
- However, Python does not have a character data type, a single character is simply a string with a length of 1.
- Square brackets can be used to access elements of the string.

Example: Get the character at position 1
(remember that the first character has the position 0):

`a = "Hello, World!"`
`print(a[1])`

Output: e

Example: String Slicing

`b = "Hello, World!"`
`print(b[2:5])`

Output: llo

- Slicing:** You can return a range of characters by using the slice syntax.
- Specify the **start** index and the **end** index (*excluded*), separated by a colon, to return a part of the string.

Functions

Loops, Sequence Operations

- Formally, a function can be defined as follows:

def <name>(<formal-parameters>):
 <body>

- The <name> of a function should be an identifier and <formal-parameters> is a (possibly empty) list of variable names (also identifiers)
- <formal-parameters> and all *local* variables declared in a function are *only* accessible in the <body> of this function
- Variables with *identical* names declared elsewhere in a program are distinct from <formal-parameters> and local variables inside a function's <body>

- Consider the following code

```
print("Happy birthday to you!")
print("Happy birthday to you!")
print("Happy birthday, dear Fred")
print("Happy birthday to you!")
```

Can we write this program with
ONLY two prints?

```
def happy():
    print("Happy birthday to you!")

def singFred():
    happy()
    happy()
    print("Happy birthday, dear Fred")
    happy()

singFred()
```

More **modular & maintainable**– changing anything in the lyric “Happy birthday to you!” requires making a change at only one place in happy(); thanks to the happy function!

EXERCISE:

- What is error in following python code:

```
word=input("Please enter two words: ")
word1=word
number=len(word)
space='/'
for i in range(number):
    if(word[i]==" "):
        word1[i]=space
print(word1)
```

- The above python code is used to first input two strings. Then it traverse for space between word and replace the space with '/' and print the new string.

You are required to write correct code for the problem.

3. What is the error in following code:

```
string="This university is best for CS aspirants"  
s=string.split()[::-1]  
l=[]  
for i in s:  
    l.append(i)  
print(" ".join[l])
```

4. Correct the error and then tell what will be the output of the code:

- a) reverse of the string
- b) aspirants CS for best is university This
- c) aspirantsCSforbestisuniversityThis
- d) error

5. What is the output of following code:

```
s1="Abc"  
s2="Xyz"  
s1_length=len(s1)  
s2_length=len(s2)  
length=s1_length if s1_length>s2_length else s2_length  
result=""  
s2=s2[::-1]  
for i in range(length):  
    if i < s1_length:  
        result=result+s1[i]  
    if i<s2_length:  
        result=result+s2[i]  
print(result)
```

- a) AXbycz
- b) AzbycX
- c) cbAzyX
- d) None of the above

6. What does following code do:

```
import re
txt="The rain in Spain"
x=re.search("^The.*Spain$",txt)
if x:
    print("Yes!We have a match")
else:
    print("No match")
```

7. What is the output of following code:

```
import re
txt="The rain in Spain"
x=re.findall("ai",txt)
print(x)
x=re.findall("portugal",txt)
print(x)
x=re.search("\s",txt)
print(x.start())
x=re.split("\s",txt)
print(x)
```

8. What is the output of the code. Explain each method.

```
import re
txt="The rain in Spain"
x=re.split("\s",txt,1)
print(x)
x=re.sub("\s","9",txt)
print(x)
x=re.sub("\s","9",txt,2)
print(x)
x=re.search("ai",txt)
print(x)
```

9. What is the error in the following code:

```
import re
txt="The rain in Spain"
x=re.search(r"\bS\w+",txt)
print(x.span())
print(x.string())
print(x.group())
```

10. Correct the above code, and then print the output.
11. What is the output of following code:

```
def fn(x):  
    try:  
        print(a/b)  
    except TypeError:  
        print('Unsupported operation')  
    except ZeroDivisionError:  
        print('Divisor by zero not allowed')  
fn(0)
```

- a) Divisor by zero not allowed
- b) Syntax Error
- c) Value Error
- d) Type Error
12. What is the difference between Arguments and parameters in python?
13. What is the error in following python code:

```
def my_function(fname,lname):  
    print(fname+" "+lname)  
my_function("Emili")
```

14. What is the output of following code:

```
if 5>2:  
    print("Five is greater than two!")  
if 5>2:  
    print("Five is greater than two!")
```

- a) Five is greater than two!
Five is greater than two!
- b) Indentation error
- c) Unindent does not match any outer indentation level
- d) Five is greater than two!
Five is greater than two!

15. What is the error in the following code. Correct the error and rewrite the code to print maximum out of three numbers.

```
def max(x,y,z):
    if(x>y):
        if(x>z):
            return x
        else:
            return z
    else:
        if(y>z):
            return y
        else:
            return z
a=int(input("Enter first number: "))
b=int(input("Enter second number: "))
c=int(input("Enter third number: "))
print("Finding the maximum out of a:",a,"and b:",b,"and c:",c)
d=max(a,b,c)
print(d,"is maximum")
```

Bonus Question:

16. Write a python code to find frequency of maximum occurring character in a python string.